CHAPTER 12

Parametric representations and Cartesian equations

S65, S67, S70

Objectives

Describe a line and a plane; determine intersections; choose a method for a projection.

Course notes

Throughout this chapter, coordinates are given in an orthonormal frame of space. This assumption is necessary to identify a normal vector from the coefficients of an equation and to calculate distances.

Parametrised lines

The line through $A(x_A, y_A, z_A)$ with non-zero direction vector (a, b, c) has the representation

$$d : \begin{cases} x = x_A + at \\ y = y_A + bt \\ z = z_A + ct \end{cases} \qquad (t \in \mathbb{R}).$$

A point belongs to the line if there exists *one and the same* real number t satisfying the three equalities. A line has many representations: change the starting point or multiply the direction vector by a non-zero real number.

Example. $(x, y, z) = (1, 2, 0) + t(2, -1, 3)$ describes the same line as $(3, 1, 3) + s(4, -2, 6)$; the link is $t = 1 + 2s$.

Cartesian planes

Equation of a plane

If $(a, b, c) \neq (0, 0, 0)$, the set $ax + by + cz + d = 0$ is a plane with normal vector (a, b, c). The plane with normal $\vec{n}$ through A is constructed by writing $\overrightarrow{AM} \cdot \vec{n} = 0$.

Proof. The condition $\overrightarrow{AM} \cdot \vec{n} = 0$ is written $a(x - x_A) + b(y - y_A) + c(z - z_A) = 0$. Expanding, $d = -ax_A - by_A - cz_A$. Conversely, every point satisfying this equation satisfies the orthogonality condition, hence belongs to the plane.

Example. The plane with normal $(2, -1, 3)$ through $A(1, 2, 0)$ is $2(x - 1) - (y - 2) + 3z = 0$, that is $2x - y + 3z = 0$.

Intersections

For a line and a plane, replace x, y, z in the equation of the plane by their parametric expressions. A solution in t gives a single intersection point; an identity gives a line contained in the plane;

a contradiction gives a parallel line that is disjoint from the plane. For two planes, non-collinear normals give an intersection along a line; solve the system of two equations by choosing one variable as a parameter. If the normals are collinear, the planes are parallel: distinguish coincident from disjoint. For two lines, use independent parameters s, t and solve the three equations. With no solution and non-collinear direction vectors, the lines are not coplanar.

Projection onto a plane

If $\mathcal{P} : ax + by + cz + d = 0$, look for the orthogonal projection of M onto the normal line through M: $H = M + t(a, b, c)$. Substitute into the equation of the plane to determine t, then calculate MH. We deduce, without making it a formula to be learned in isolation,

$$d(M, \mathcal{P}) = \frac{|ax_M + by_M + cz_M + d|}{\sqrt{a^2 + b^2 + c^2}}.$$

Essential check. After calculating a projection, check simultaneously that $H \in \mathcal{P}$ and that $\overrightarrow{MH}$ is collinear with the normal.

Research activity

Three lines facing a plane — 30 min.

In an orthonormal frame of space, consider the plane $\mathcal{P}$ with equation $x + y + z = 3$. In each question, the parameter t ranges over $\mathbb{R}$.

1. The line d_1 has the representation $(x; y; z) = (t; t; t)$. Determine $d_1 \cap \mathcal{P}$.
2. The line d_2 has the representation $(x; y; z) = (3 + t; -t; 0)$. Determine $d_2 \cap \mathcal{P}$.
3. The line d_3 has the representation $(x; y; z) = (t; -t; 0)$. Determine $d_3 \cap \mathcal{P}$.
4. For each line, state whether the equation obtained after substitution into that of the plane has a unique solution, no solution, or every real number as a solution. Link these cases to the geometric positions found.

Progressive exercises

E1 · Practice. In a frame of space, consider $A(1; -1; 2)$ and $B(3; 0; 1)$. Give a parametric representation of the line (AB).

E2 · Practice. In an orthonormal frame, determine a Cartesian equation of the plane through $A(1; 2; 3)$ with normal vector $\vec{n} = (1; 0; -2)$.

E3 · Practice. In a frame of space, the line d is given by $(x; y; z) = (t; 1 + t; 2 - t)$, $t \in \mathbb{R}$, and the plane $\mathcal{P}$ by $x + 2y + z = 8$. Determine their intersection.

E4 · Practice. In a frame of space, determine the intersection of the planes with equations $x + y + z = 1$ and $x - y = 0$. Give a parametric representation if the intersection is a line.

E5 · Practice. In a frame of space, study the relative position of the planes with equations $2x - y + z = 3$ and $4x - 2y + 2z = 5$. Justify.

Textbook or past-paper work

Baccalauréat, Metropolitan France, 17 June 2025: exercise 3, statements 1 to 4 (p. 5–6). At home after S70. Justify each answer; for a false statement, explain precisely what refutes it. The paper is available here: official past paper. Use the data and instructions of this past paper.

Homework

Homework 12 — Projection and intersection — 55 min.

In an orthonormal frame of space, consider the point $M(3;0;0)$ and the planes $\mathcal{P} : x + 2y + 2z = 9$ and $\mathcal{Q} : x - y = 0$.

1. Give the coordinates of a non-zero normal vector to $\mathcal{P}$.
2. Determine the coordinates of the orthogonal projection H of M onto $\mathcal{P}$, then calculate the distance from M to $\mathcal{P}$.
3. Show that $\mathcal{P} \cap \mathcal{Q}$ is a line and give a parametric representation of it.
4. Does the point H belong to this line? Justify.
5. Determine the coordinates of the point K of this line whose third coordinate (the "cote") is zero. Calculate MK, then compare MK and MH.

Teaching notes and solutions

Teacher notes

Use two different parameters to compare two lines. Check all three coordinates, even if two suffice to calculate candidates. A Cartesian equation on its own generally describes a plane in space; it does not describe a line. Reuse the projection from Chapter 11 to give meaning to the solving of systems.

Solutions to the activity

1. A point of d_1 has coordinates (t,t,t). It belongs to the plane if and only if $3t=3$, that is $t=1$. The intersection is therefore the single point $(1,1,1)$.
2. For a point of d_2, $x+y+z=3+t-t+0=3$ for every t. All points of the line are in the plane: $d_2 \subset \mathcal{P}$.
3. For a point of d_3, $x+y+z=t-t+0=0$. The equation $0=3$ has no solution: the line is disjoint from the plane. Its direction vector $(1,-1,0)$ is orthogonal to the normal $(1,1,1)$; it is parallel to the plane.
4. After substitution, we obtain respectively an equation with a single solution, an identity true for every parameter value, and a contradiction. These three cases correspond to a single intersection point, containment, and no intersection.

E1. $\overrightarrow{AB}=(2,1,-1)\neq\vec{0}$. The line therefore has the representation $(x,y,z)=(1,-1,2)+t(2,1,-1)$, $t\in\mathbb{R}$. The values $t=0$ and $t=1$ give back A and B.

E2. For $A(1,2,3)$, the condition $\overrightarrow{AM}\cdot(1,0,-2)=0$ is $(x-1)-2(z-3)=0$. After expanding, the plane has equation $x-2z+5=0$. The point A satisfies it.

E3. After substitution, $t+2(1+t)+(2-t)=8$, that is $2t+4=8$, so $t=2$. The unique intersection is $(2,3,0)$; we check $2+2\times3+0=8$.

E4. The second equation gives $x=y$. Let $x=y=t$; the first becomes $2t+z=1$. The set of solutions is $(x,y,z)=(t,t,1-2t)=(0,0,1)+t(1,1,-2)$, a line.

E5. The normals are collinear, the second being twice the first. But doubling $2x-y+z=3$ gives $4x-2y+2z=6$, which is incompatible with the right-hand side 5. The planes are therefore strictly parallel, not intersecting.

Solutions to Homework 12

1. The coefficients of x, y, z give the non-zero normal $\vec{n} = (1, 2, 2)$ in the orthonormal frame.
2. Look for H on the normal line through M: $H = (3 + t, 2t, 2t)$. Membership of $\mathcal{P}$ gives $3 + t + 4t + 4t = 9$, so $t = \frac{2}{3}$. Thus $H = (\frac{11}{3}, \frac{4}{3}, \frac{4}{3})$. We check $\frac{11}{3} + \frac{8}{3} + \frac{8}{3} = 9$ and $\overrightarrow{MH} = (\frac{2}{3})\vec{n}$. So H is indeed the orthogonal projection, and $MH = (\frac{2}{3})\sqrt{1 + 4 + 4} = 2$.
3. The normals $(1, 2, 2)$ and $(1, -1, 0)$ are not collinear: the planes meet along a line. Since $x = y$, let $x = y = s$. The first equation gives $3s + 2z = 9$, that is $z = \frac{9-3s}{2}$. The intersection is the line $(x, y, z) = (s, s, \frac{9-3s}{2})$, $s \in \mathbb{R}$.
4. The point H has $x_H = \frac{11}{3} \neq \frac{4}{3} = y_H$, so it belongs neither to $\mathcal{Q}$ nor to the line of intersection.
5. The cote is the z-coordinate. Setting it to zero requires $9 - 3s = 0$, so $s = 3$ and $K = (3, 3, 0)$. Then $MK = 3 > 2 = MH$. This result is consistent with the property of the orthogonal projection: among all points of the plane, H is the closest to M.

Solutions to the textbook or past-paper work

1. True: the line passes through B and its direction vector $(-2; -1; 3)$ equals $-\frac{1}{2}\overrightarrow{AB}$.
2. False: $\vec{n} \cdot \overrightarrow{OB} = 15 - 4 - 1 = 10 \neq 0$.
3. False: for $k = -4$ and $s = \frac{5}{2}$, the two representations give $(11; 12; -14)$. The lines intersect, so they are coplanar.
4. True: $d(C, \mathcal{P}) = \frac{|2-(-1)+2+1|}{\sqrt{1+1+1}} = 2\sqrt{3}$.